Statistical Code Review

1. Summary Metrics

Metric	Result
Cyclomatic Complexity	**809**
Cognitive Complexity	**426**
Open Issues	**195**
Maintainability Issues	**177**
Reliability Issues	**18**
Code Duplication	**2.2%**
Test Coverage	**0.0%**
Quality Gate	**Passed**

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2. Key Findings & Interpretation

- The project exhibits **high cyclomatic and cognitive complexity**, mainly in controller classes, indicating complex and hard-to-maintain logic.
- A **large number of maintainability issues** suggests the need for refactoring large methods and improving separation of concerns.
- **Code duplication is relatively low (2.2%)**, showing acceptable reuse practices.
- No critical **security issues** were detected.
- Unit test coverage is 0%, as tests were skipped due to failing test cases unrelated to static analysis.

3. High-Priority Issues

#	Severity	Message	File	Line
1	BLOCKER	'password' detected in this expression, review this		
		potentially hard-coded password.	src/test/java/model/	128
2	BLOCKER	'password' detected in this expression, review this		
		potentially hard-coded password.	src/test/java/model/	99
3	CRITICAL	Define a constant instead of duplicating this literal		
		"togglePassword" 3 times.	src/test/java/contro	106
4	CRITICAL	Define a constant instead of duplicating this literal		
		"currentPwdField" 5 times.	src/test/java/contro	33
5	CRITICAL	Define a constant instead of duplicating this literal		
		"Central Processing Unit" 4 times.	src/test/java/model/	72
6	CRITICAL	Define a constant instead of duplicating this literal		
		"userDao" 3 times.	src/test/java/contro	49
7	CRITICAL	Define a constant instead of duplicating this literal		
		"moreMenu" 4 times.	src/test/java/contro	168
8	CRITICAL	Define a constant instead of duplicating this literal		

		"Subject-" 3 times.	src/test/java/model/	47
9	CRITICAL	Define a constant instead of duplicating this literal		
		"oldpass" 4 times.	src/test/java/contro	43
10	CRITICAL	Define a constant instead of duplicating this literal		
		"classCodeField" 4 times.	src/test/java/contro	72
11	CRITICAL	Define a constant instead of duplicating this literal		
		"passwordField" 5 times.	src/test/java/contro	42
12	CRITICAL	Define a constant instead of duplicating this literal		
		"teacherAddClass" 3 times.	src/test/java/contro	76
13	CRITICAL	Define a constant instead of duplicating this literal		
		"definitionPane" 4 times.	src/test/java/contro	25
14	CRITICAL	Define a constant instead of duplicating this literal		
		"passwordTextField" 4 times.	src/test/java/contro	38
15	CRITICAL	Define a constant instead of duplicating this literal		
		"newPwdField" 5 times.	src/test/java/contro	34
16	CRITICAL	Refactor this method to reduce its Cognitive		
		Complexity from 20 to the 15 allowed.	src/test/java/model/	203
17	CRITICAL	Define a constant instead of duplicating this literal		
		"subjectField" 3 times.	src/test/java/contro	74

18	CRITICAL	Add a nested comment explaining why this method is empty, throw an UnsupportedOperationException or complete the implementation.	src/test/java/contro	35
19	CRITICAL	Define a constant instead of duplicating this literal "newpassword" 3 times.	src/test/java/contro	148
20	CRITICAL	Define a constant instead of duplicating this literal "emailField" 3 times.	src/test/java/contro	41
21	CRITICAL	Define a constant instead of duplicating this literal "-fx-font-family: 'Noto Sans Lao';" 4 times.	src/main/java/contro	37
22	CRITICAL	Define a constant instead of duplicating this literal "onSave" 4 times.	src/test/java/contro	99
23	CRITICAL	Define a constant instead of duplicating this literal "CreatorTerm-" 3 times.	src/test/java/model/	204
24	CRITICAL	Define a constant instead of duplicating this literal "classBtn" 4 times.	src/test/java/contro	51
25	CRITICAL	Define a constant instead of duplicating this literal "Term-" 3 times.	src/test/java/model/	85
26	CRITICAL	Define a constant instead of duplicating this literal "123456" 3 times.	src/test/java/contro	96

27	CRITICAL	Add a nested comment explaining why this method is empty, throw an UnsupportedOperationException or complete the implementation.	src/test/java/contro	36
28	CRITICAL	Define a constant instead of duplicating this literal "onAdd" 3 times.	src/test/java/contro	124
29	CRITICAL	Define a constant instead of duplicating this literal "classNameLabel" 3 times.	src/test/java/contro	27
30	CRITICAL	Define a constant instead of duplicating this literal "quizTitleLabel" 3 times.	src/test/java/contro	23

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4. Recommendations

☐ **1. [BLOCKER] 'password' detected in this expression, review this potentially hard-coded password.**

→ Recommendation: Fix immediately. This issue may break functionality or cause critical failures.

☐ **2. [BLOCKER] 'password' detected in this expression, review this potentially hard-coded password.**

→ Recommendation: Fix immediately. This issue may break functionality or cause critical failures.

☐ **3. [CRITICAL] Define a constant instead of duplicating this literal "togglePassword" 3 times.**

→ Recommendation: Prioritize fixing this issue. Follow the rule description in SonarQube.

☐ **4. [CRITICAL] Define a constant instead of duplicating this literal "currentPwdField" 5 times.**

→ Recommendation: Prioritize fixing this issue. Follow the rule description in SonarQube.

☐ **5. [CRITICAL] Define a constant instead of duplicating this literal "Central Processing Unit" 4 time**

→ Recommendation: Prioritize fixing this issue. Follow the rule description in SonarQube.

☐ **6. [CRITICAL] Define a constant instead of duplicating this literal "userDao" 3 times.**

→ Recommendation: Prioritize fixing this issue. Follow the rule description in SonarQube.

☐ **7. [CRITICAL] Define a constant instead of duplicating this literal "moreMenu" 4 times.**

→ Recommendation: Prioritize fixing this issue. Follow the rule description in SonarQube.

☐ **8. [CRITICAL] Define a constant instead of duplicating this literal "Subject-" 3 times.**

→ Recommendation: Prioritize fixing this issue. Follow the rule description in SonarQube.

☐ **9. [CRITICAL] Define a constant instead of duplicating this literal "oldpass" 4 times.**

→ Recommendation: Prioritize fixing this issue. Follow the rule description in SonarQube.

☐ **10. [CRITICAL] Define a constant instead of duplicating this literal "classCodeField" 4 times.**

→ Recommendation: Prioritize fixing this issue. Follow the rule description in SonarQube.

☐ **11. [CRITICAL] Define a constant instead of duplicating this literal "passwordField" 5 times.**

→ Recommendation: Prioritize fixing this issue. Follow the rule description in SonarQube.

☐ **12. [CRITICAL] Define a constant instead of duplicating this literal "teacherAddClass" 3 times.**

→ Recommendation: Prioritize fixing this issue. Follow the rule description in SonarQube.

☐ **13. [CRITICAL] Define a constant instead of duplicating this literal "definitionPane" 4 times.**

→ Recommendation: Prioritize fixing this issue. Follow the rule description in SonarQube.

☐ **14. [CRITICAL] Define a constant instead of duplicating this literal "passwordTextField" 4 times.**

→ Recommendation: Prioritize fixing this issue. Follow the rule description in SonarQube.

☐ **15. [CRITICAL] Define a constant instead of duplicating this literal "newPwdField" 5 times.**

→ Recommendation: Prioritize fixing this issue. Follow the rule description in SonarQube.

☐ **16. [CRITICAL] Refactor this method to reduce its Cognitive Complexity from 20 to the 15 allowed.**

→ Recommendation: Refactor the method into smaller units to reduce complexity.

☐ **17. [CRITICAL] Define a constant instead of duplicating this literal "subjectField" 3 times.**

→ Recommendation: Prioritize fixing this issue. Follow the rule description in SonarQube.

☐ **18. [CRITICAL] Add a nested comment explaining why this method is empty, throw an UnsupportedO**

→ Recommendation: Prioritize fixing this issue. Follow the rule description in SonarQube.

☐ **19. [CRITICAL] Define a constant instead of duplicating this literal "newpassword" 3 times.**

→ Recommendation: Prioritize fixing this issue. Follow the rule description in SonarQube.

☐ **20. [CRITICAL] Define a constant instead of duplicating this literal "emailField" 3 times.**

→ Recommendation: Prioritize fixing this issue. Follow the rule description in SonarQube.

☐ **21. [CRITICAL] Define a constant instead of duplicating this literal "-fx-font-family: 'Noto Sans La** → Recommendation: Prioritize fixing this issue. Follow the rule description in SonarQube.

☐ **22. [CRITICAL] Define a constant instead of duplicating this literal "onSave" 4 times.**

→ Recommendation: Prioritize fixing this issue. Follow the rule description in SonarQube.

☐ **23. [CRITICAL] Define a constant instead of duplicating this literal "CreatorTerm-" 3 times.**

→ Recommendation: Prioritize fixing this issue. Follow the rule description in SonarQube.

☐ **24. [CRITICAL] Define a constant instead of duplicating this literal "classBtn" 4 times.**

→ Recommendation: Prioritize fixing this issue. Follow the rule description in SonarQube.

☐ **25. [CRITICAL] Define a constant instead of duplicating this literal "Term-" 3 times.**

→ Recommendation: Prioritize fixing this issue. Follow the rule description in SonarQube.

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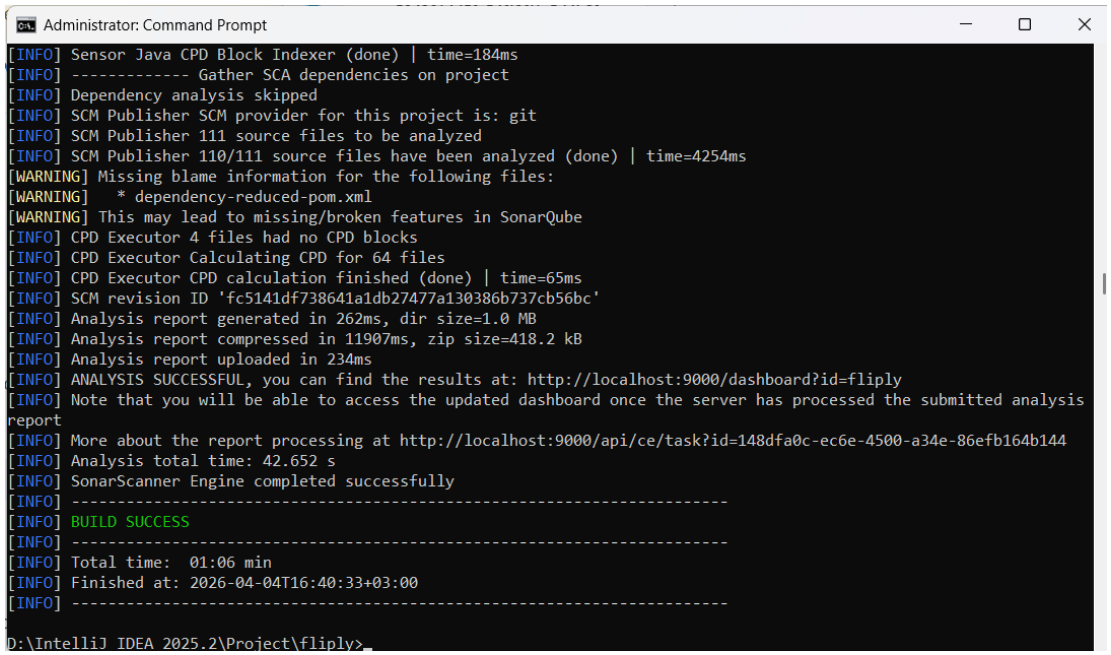
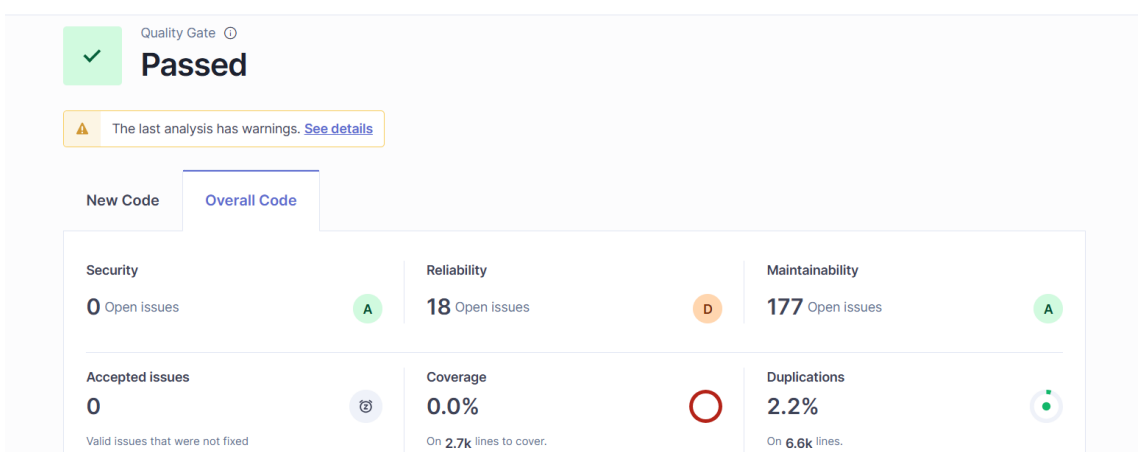
5. Visual Evidence

The following visual evidence is included as screenshots:

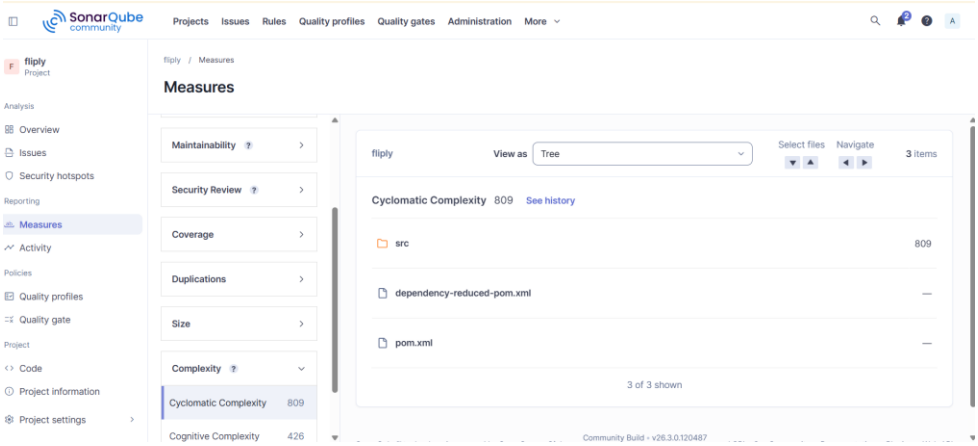
- SonarQube analysis completion (BUILD SUCCESS) :

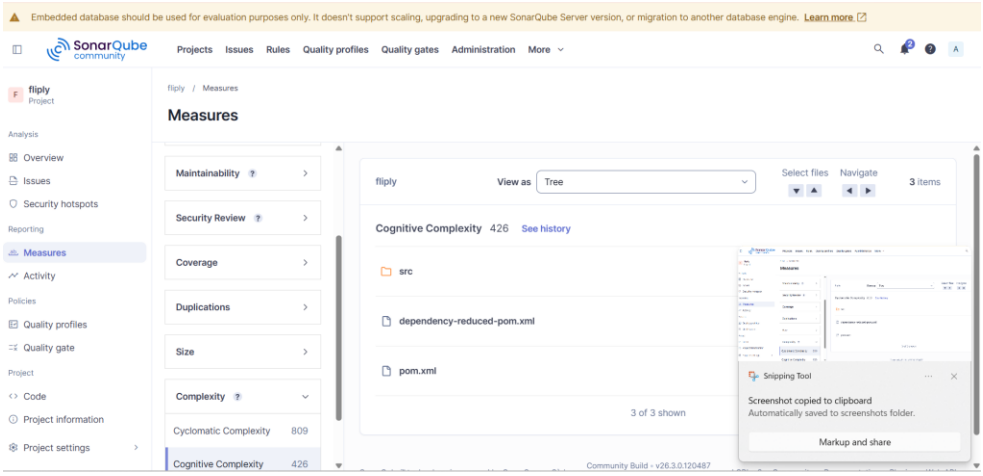


- Project Dashboard (overall metrics)

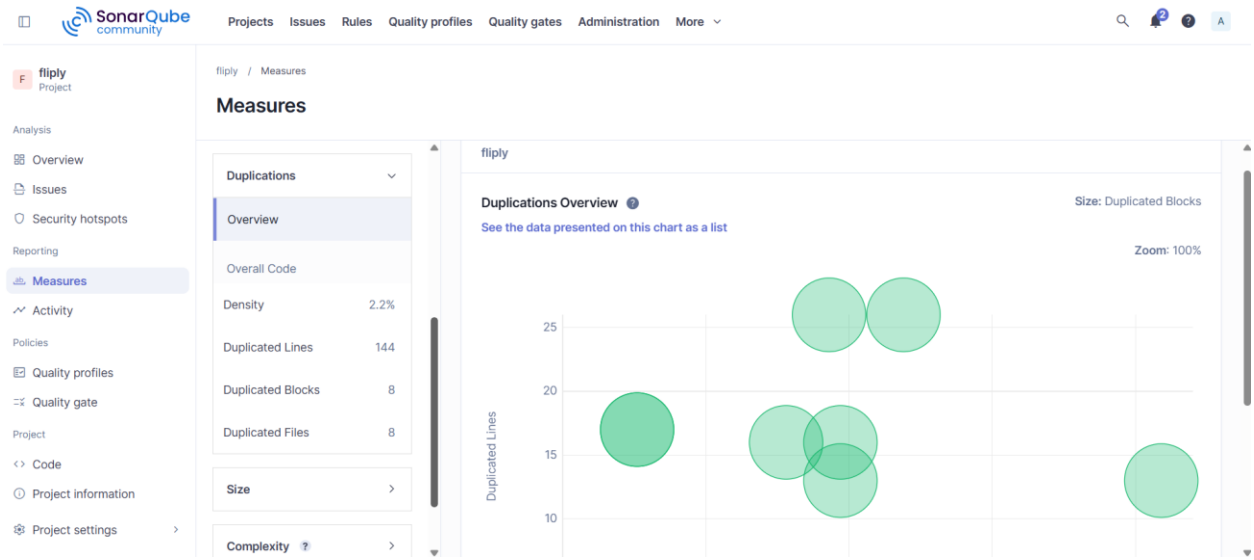
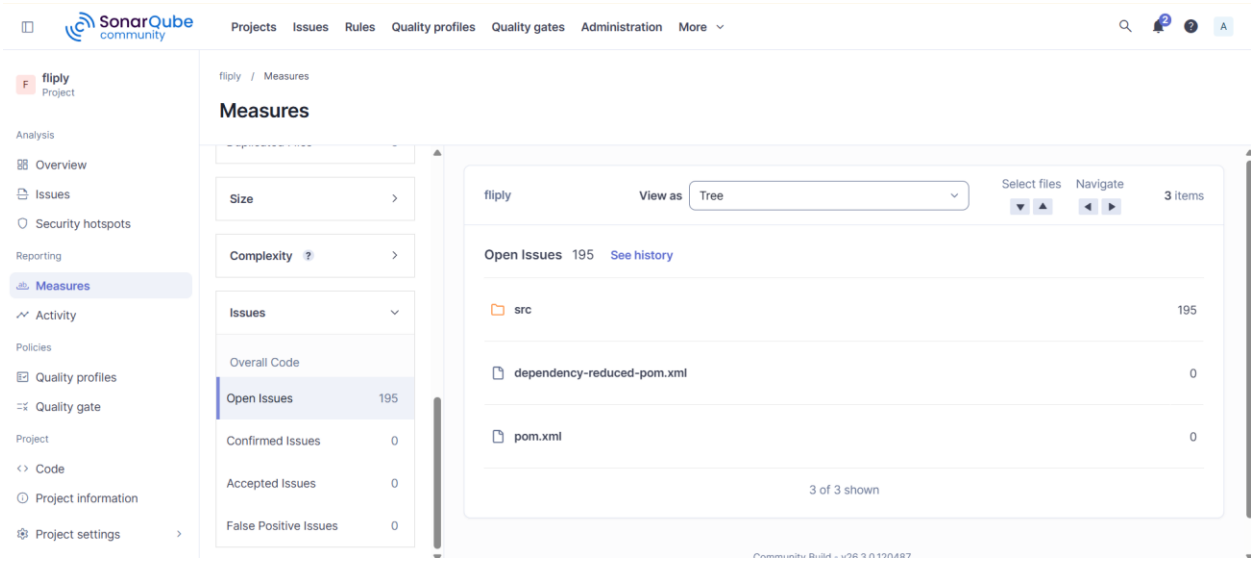


- Complexity metrics (Cyclomatic & Cognitive Complexity)





- Duplicate code chart



- Issue summary and Quality Gate status